

Temporal proximity affects the detailed recall of events in a nested hierarchy

Bryan Hong^{1*}, My An Tran¹, Heidi Cheng¹, Yinqi Huang¹, Morgan D. Barense^{1,2}

¹ Department of Psychology, University of Toronto, Toronto, ON, Canada

² Rotman Research Institute, Baycrest Hospital, Toronto, ON, Canada

***Corresponding author:**

Bryan Hong

100 St. George Street (Department of Psychology, Fourth Floor)

Toronto, ON, CA M5S 3G3

bryan.hong@mail.utoronto.ca

Abstract: 246 words **Main text:** 7299 words

Acknowledgements: This work was supported by a Scholar Award from the James S. McDonnell Foundation and a Natural Sciences and Engineering Research Council (NSERC) Discovery grant. M.D.B. is also supported by a Canada Research Chair and a Max and Gianna Glassman Chair in Neuropsychology. B.H. is also supported by an NSERC Postgraduate Scholarships – Doctoral award and the Ontario Graduate Scholarship. We thank Brian Levine for sharing documentation for administering and scoring the Autobiographical Interview, and Aya Ben-Yakov for sharing data on the event boundary ratings for the movie *Forrest Gump*.

Data availability: Data from this work is available as an open-access dataset:
<https://osf.io/yfn6m/>

CrediT author statement: Bryan Hong: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. My An Tran: Investigation. Heidi Cheng: Investigation. Yinqi Huang: Investigation. Morgan D. Barense: Conceptualization, Methodology, Writing – review & editing, Supervision, Funding acquisition.

Abstract

Retrieved-context models of memory posit that a slowly drifting neural context signal is encoded alongside items in memory, linking them together in time. However, this signal can be disrupted by event boundaries, affecting the temporal relationships between items across event boundaries. Here, we investigated how temporal proximity and event boundaries affect the detailed recall of events. Specifically, we assessed memory for events in the movie *Forrest Gump*, chosen specifically because of its hierarchical event structure with individual events in the movie being nested within broader life-periods (e.g., childhood, college, etc.). Importantly, the trial order at recall was structured to manipulate the similarity between adjacently cued events, with events tested on adjacent trials being (1) either temporally close or far and (2) either within or across a life period. Participants provided both more event-specific details and more false memories for a cued event when the preceding event at test came from the same life period, suggesting reinstatement of a shared schematic representation to scaffold recall. Further, participants provided more intrusions from non-cued events after just recalling an event that was temporally distant, with this effect exacerbated if the preceding event was also across a life period boundary. We found that these intrusions tended to come from events that were similar to the cued event, suggesting that participants may be using this supplementary information to support recall. Altogether, these results demonstrate how temporal proximity and event boundaries interact to shape how we recollect events within a broader narrative.

Keywords: episodic memory, context, event boundaries, schema, detailed recall, narratives

Public significance statement

How is our memory for a given moment shaped by its associations to what we just recalled last? Previous work has demonstrated a crucial role of time in guiding memory search, linking together items that take place close together in time. However, these temporal associations can be disrupted, particularly at event boundaries, which allow us to segment our continuous experience into discrete events. In the current study, we demonstrated how temporal associations and event boundaries affect memory for naturalistic events, where time unfolds continuously alongside other dimensions of experience. Specifically, we tested memory for events in the film *Forrest Gump*, chosen specifically it consists of individual events nested within broader life periods. We found that the amount and types of details that participants recalled was affected by both the temporal proximity (i.e., how close together events were in time) and the life period congruency (i.e., whether events shared a life period) of consecutively cued events. These findings demonstrate how temporal structure and event segmentation shape our ability to recollect real-world experiences.

Temporal proximity affects the detailed recall of events in a nested hierarchy

As we traverse our experience of life, we are continually moving forward in time—this intimate tie between time and our experience naturally extends to the formation of episodic memories, which allows us to embed an event at its given point in time and space (Tulving, 1972, 2002). This defining feature of episodic memory makes time a critical dimension that we can use to organize our memories. We can use time to help sort memories for different events, with events that occur nearby in time thought to be more similar than those that occur further away in time. These temporal connections can link different memories together, allowing the retrieval of one event to facilitate the retrieval of another that occurred at a nearby timepoint.

The function of time in the organization of memory has been captured by retrieved context models, a family of computational models put forward to explain the organization of search through episodic memory (Howard & Kahana, 2002; Kahana, 2020; Polyn et al., 2009a; Polyn & Cutler, 2017). A hallmark feature of these models is a gradually changing representation of context that is encoded alongside a given event—the retrieval of a given event simultaneously activates its associated contextual representation from initial encoding, which is then used to guide subsequent memory search (Kahana, 2020; Polyn & Cutler, 2017). These models have been able to account for several phenomena that are consistently observed in the free recall of word lists, such as the *temporal contiguity effect*, where participants show a tendency to recall items in an order that is similar to how they were initially encoded (Healey et al., 2018; Kahana, 1996).

Despite being initially used to explain the dynamics of free recall for word lists, retrieved context models have been used to explain similar patterns of recall in more naturalistic stimuli (Healey et al., 2018). For example, temporal contiguity effects have been observed for the recall

of events encoded outside of a laboratory setting, such as news stories (Uitvlugt & Healey, 2018) and autobiographical events (Moreton & Ward, 2010). In addition to the relative complexity of these stimuli compared to word lists, these findings suggest that the predictions from retrieved context models can scale from the intervals observed in typical word list studies (e.g., seconds) up to those observed between events in our typical day-to-day experience (e.g., days, months, years). These results support hypotheses proposing that our internal representation of time is scale-invariant, so that these temporal relationships can be maintained regardless of the unit of time (Howard et al., 2015; Shankar & Howard, 2012).

In addition to the well-established role of time in the organization of memory search, findings suggest that the degree of temporal reinstatement from encoding at recall is positively linked with episodic detail. For example, this has been captured in word list free recall tasks by asking participants to make Remember/Know judgements at recall, which are used to capture the degree of subjective re-experiencing of an event (Tulving, 1985). Items with a “Remember” judgement tend to be accompanied with stronger temporal reinstatement, as quantified by the size of the temporal contiguity effect, compared to items with a “Know” judgement (Sadeh et al., 2015). Similarly, participants demonstrate better performance in recognition memory tasks when items are tested in an order that matches the initial encoding order (Averell et al., 2016; Schwartz et al., 2005; Stern et al., 2020). This relationship has also been demonstrated with free recall for both naturally occurring (Pathman et al., 2023) and staged naturalistic events (Diamond & Levine, 2020), where there is a positive correlation between the size of the temporal contiguity effect and the amount of episodic detail provided. Together, these findings suggest that the reinstatement of temporal context from encoding can act as a scaffold to facilitate retrieval of episodic detail at subsequent recall.

However, a growing body of work from the event segmentation literature has demonstrated instances in which our internal representation of time can be altered (Clewett et al., 2019; DuBrow et al., 2017). Event segmentation theory posits that we form and maintain active event representations as we traverse through our experience of the world (DuBrow, 2024; Radvansky & Zacks, 2017; Zacks et al., 2007). This event representation incorporates ongoing perceptual and sensory input with a relevant *event schema*, encompassing our prior knowledge of the current situation (Baldassano et al., 2018; Gilboa & Marlatte, 2017). This allows us to form predictions for the ongoing event—for example, if we are attending a lecture in a classroom, we may expect to see desks, a blackboard, and an instructor at the front of the room. When an occurrence violates our expectations, an *event boundary* occurs, directing us to update our representation to match the new event (Kurby & Zacks, 2008). Returning to the previous classroom example, a sudden loud ring of the school bell may deviate from our expectations, thus signalling that a new event is unfolding. These event boundaries have been shown to disrupt the perceived temporal relationships between items in memory. For example, event boundaries can result in a perceived dilation in time, with items being perceived as occurring further apart in time if they take place across different events than if they take place within the same event (Brunec et al., 2017; Ezzyat & Davachi, 2011, 2014; Faber & Gennari, 2015). In addition, temporal order memory is better for items that occur within the same event compared to those that span different events (DuBrow & Davachi, 2013; Heusser et al., 2018). However, many of these studies typically involve tasks using laboratory-based stimuli, such as words or object images, where events are defined by an overlapping feature, such as shared category membership, or an associated colour background or image (DuBrow & Davachi, 2016; Ezzyat & Davachi, 2014; Heusser et al., 2018). Memory for a given item is then usually assessed using a

binary metric (e.g., “recalled” vs. “not recalled”), making it difficult to capture the subjective re-experiencing of a given item.

We aimed to extend this work by investigating how temporal context and event boundaries interact for the detailed recall of events within a broader narrative, taking an ecological approach to probe the influence of time as it unfolds with other dimensions of naturalistic experience (Lee et al., 2020). To do this, we leveraged the hierarchical nature of event segmentation, whereby segmentation can occur at various levels of granularity—in other words, shorter fine-grained events (e.g., “riding the school bus”, “going to class”, “going to the principal’s office”, etc.) can be nested within longer coarse-grained events (e.g., “day at school”) (Kurby & Zacks, 2011; Zacks et al., 2001, 2007). Specifically, we tested memory for the audiovisual movie *Forrest Gump* (R. Zemeckis, Paramount Pictures, 1994), which depicts the story of the eponymous character across several decades of his life in a relatively linear narrative. Importantly, this movie was chosen because it consists of individual events (e.g., “going to the principal’s office”, “going to the doctor”, “meeting a friend on the school bus”, etc.) nested within broader life periods (e.g., “childhood”, “college”, “fighting in the Vietnam War”, etc.), allowing us to test how temporal proximity affects detailed memory for more fine-grained events both within and across these coarser life period boundaries (Figure 1A). We manipulated the trial order at recall during a cued recall task such that adjacent trials tested events that were (1) either temporally close or far in the movie and (2) either within or across a life period (Figure 1B). To investigate whether any of these effects changed after time-dependent consolidation mechanisms, we also tested participants either (1) immediately or (2) 24-hours after watching the movie (Figure 1C). We then transcribed and scored recall responses to categorize the types of details provided by participants. We hypothesized that consecutively recalling similar events

(i.e., temporally close or within life periods) would facilitate contextual reinstatement and result in richer episodic recall for the cued events. Recall of consecutive events within the same life period was also predicted to increase the number of false details generated due to the activation of a shared schematic representation. Furthermore, we hypothesized that consecutively recalling dissimilar events (i.e., temporally far or across life periods) would result in more intrusions from non-cued events. We predicted that these intrusions would originate from events that were similar to the cued event, with these details helping to scaffold memory by filling in potential gaps when there was a perceived discontinuity between tested events.

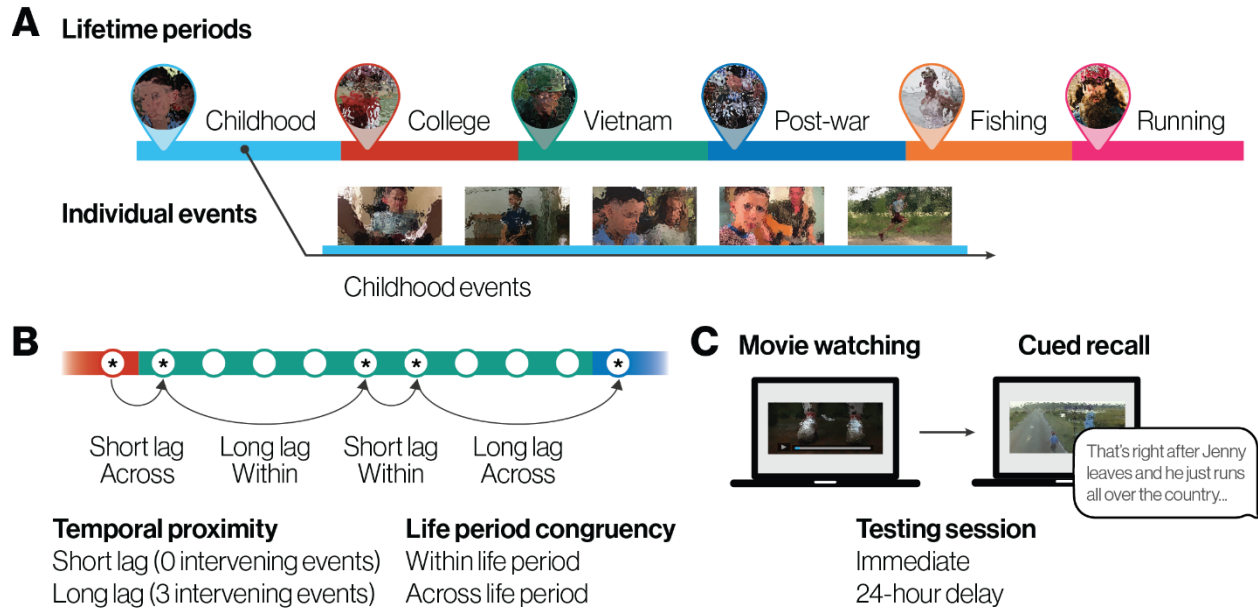


Figure 1. Schematic overview of task stimuli and design. (A) Participants were shown the movie *Forrest Gump*, which depicts the story of a man across various life periods. Each life period itself is made up of several individual events, resulting in a nested hierarchy of events in the movie. Events were defined based on event boundaries characterized in Ben-Yakov & Henson (2018). Life periods were defined by asking a separate set of individuals to group these events into “meaningful units”—life periods shown here are for demonstration purposes and were not displayed to participants. (B) At cued recall, the trial order was manipulated to vary temporal proximity and life period congruency between adjacently tested events. This schematic depicts individual events as circles on a timeline of the movie, with each coloured section on the timeline representing different life periods—circles with an asterisk denote events that are cued during the recall task. Adjacent events could either be a short lag away, with 0 intervening events, or a long lag away, with 3 intervening events. In addition, adjacent events could either be within or across a life period boundary. All events were tested in chronological order as they were seen in the movie. (C) To investigate potential effects of time-dependent consolidation mechanisms, participants were randomly assigned to complete the cued recall task either immediately or 24 hours after watching the movie. Images are blurred in this figure for copyright reasons—participants were presented with unblurred images in the study.

Methods

Participants

61 participants were recruited from the University of Toronto community and randomly assigned to complete the cued recall task immediately after ($n = 30$, $M_{\text{Age}} = 22.83$ years, $SD_{\text{Age}} = 2.26$ years, $\text{Range}_{\text{Age}} = 19\text{-}27$ years, 7 men/23 women) or 24 hours after ($n = 31$, $M_{\text{Age}} = 22.61$ years, $SD_{\text{Age}} = 2.08$ years, $\text{Range}_{\text{Age}} = 19\text{-}30$ years, 9 men/22 women) watching the movie. All participants had normal or corrected-to-normal vision, and no reported history of psychological or neurological disorders, or brain damage (i.e., stroke or surgery). All participants provided informed consent online prior to the study and received monetary compensation for their participation. The study was approved by the Research Ethics Board at the University of Toronto.

Stimuli

In the current study, we used the audiovisual movie *Forrest Gump* (R. Zemeckis, Paramount Pictures, 1994) in English, edited to be two hours in duration according to Hanke et al. (2016)—all references to *Forrest Gump* onwards will allude to this edited version. 25 participants reported having previously seen the movie (last watched a mean of 6.31 years ago; 13 immediate testing session/12 24-hour delay testing session).

We used the boundaries identified in Ben-Yakov & Henson (2018) to segment the movie into discrete events, where participants indicated with a keypress when an event ended and a new one began, following the same event segmentation protocol as described in Zacks et al. (2010). We defined an event boundary to occur when over 50% (i.e., 9 of 16) of the participants had a matching response—this segmented the movie into 83 discrete events, ranging in duration

between 6.87 to 415.87 seconds. A representative screenshot was then manually selected for each event.

To characterize life periods, a separate group of 10 raters ($M_{\text{Age}} = 20.7$ years, $SD_{\text{Age}} = 1.34$ years, $\text{Range}_{\text{Age}} = 18\text{-}22$ years, 9 women) were asked to first watch *Forrest Gump*. Afterwards, they were presented with screenshots corresponding to all events in the movie in order simultaneously to form a timeline of the movie. Raters were then asked to group the photos into “meaningful units” that corresponded to “themes and life periods that could stand on its own”. These life periods were required to consist of contiguous events, but there was no constraint on the number of life periods identified or the number of events within a life period. Raters were also asked to name each life period and rate how easy it was to identify the boundary between each life period on a 4-point scale, with 1 corresponding to “Very difficult” and 4 corresponding to “Very easy”. The similarity in the life period boundaries identified was assessed using Dice’s coefficient, which is a ratio of the number of matching boundaries and the mean number of boundaries identified (Baldassano et al., 2017)—one rater was excluded from analysis because they had an average pairwise Dice’s coefficient more than 2 standard deviations below the mean. Raters identified between 8 and 17 life period boundaries, with an average pairwise Dice’s coefficient of 0.56. Raters indicated that it was relatively easy to form these boundaries, with an average response of 3.39. We defined a life period boundary to occur when at least 4 participants had a matching response to arrive at 11 life periods, each consisting between 1 to 25 events.

Task design

Participants were first asked to watch the movie, *Forrest Gump*. They were told that there would be a subsequent memory test, but otherwise did not complete any tasks concurrently (Figure 1B).

Participants were randomly assigned to then complete a cued recall task either immediately or 24 hours after watching the movie. Participants who completed the cued recall task 24 hours after watching the movie reported sleeping for 7.29 hours on average ($SD = 1.27$ hours, range = 4.5-10.0 hours).

During the cued recall task, participants were sequentially presented with 35 screenshots corresponding to different events in the movie. For each trial, participants were asked to first indicate how confidently they remembered the event on a 4-point scale, with 1 corresponding to “Not very confident” and 4 corresponding to “Very confident”. Afterwards, they were asked to verbally recall the event in as much detail as possible—participants were encouraged to try to speak for at least a minute per trial. Screenshots were presented on screen for one second, with a one second fixation period before and after. A row of asterisks was presented on screen along with an auditory beep to indicate that participants could begin their verbal description. Participants were not given a time limit on their recall and indicated that they were finished recalling with a key press.

Importantly, the cued events were selected to manipulate both the temporal proximity and life period congruency of adjacent events (Figure 1B). To manipulate temporal proximity, adjacently tested events could either be separated by (1) a short lag, with zero intervening events, or (2) a long lag, with three intervening events. To manipulate life period congruency, adjacently tested events could either be (1) within a life period, or (2) across a life period. Participants were randomly assigned to one of three different trial lists to vary the cued events across participants. Trial lists were constructed so that there were an equal number of short and long lag transitions across life periods. In addition, there could not be more than three events with the same combination of temporal proximity and life combination levels in a row. Events that were less

than 20 seconds in duration were not selected for this task. Critically, all events were cued sequentially in chronological order as they were seen in the movie.

Detail scoring protocol for cued recall

Verbal descriptions were manually transcribed and scored to assess episodic memory for each event. Transcriptions were parsed into individual details according to the Autobiographical Interview (Levine et al., 2002) and then categorized into one of four detail types (Figure 2). The four detail types were event-specific details, intrusions, false memories, and general details.

Event-specific details corresponded to information that was present in the cued event. These details are roughly analogous to the internal details defined in the Autobiographical Interview (Levine et al., 2002) and included information such as the characters, places, and activities that took place in a given event. *Intrusions* corresponded to information that was present in the movie, but outside of the currently cued event. To quantify any differences in the source of intrusions resulting from the manipulation of trial order, the corresponding event for each intrusion was also identified. *False memories* corresponded to details that were incorrectly remembered from the movie. *General details* corresponded to any general information that was not specific to the movie, editorializing statements, or repetitions of previously mentioned details in the trial, similar to the external details from the Autobiographical Interview (Levine et al., 2002).

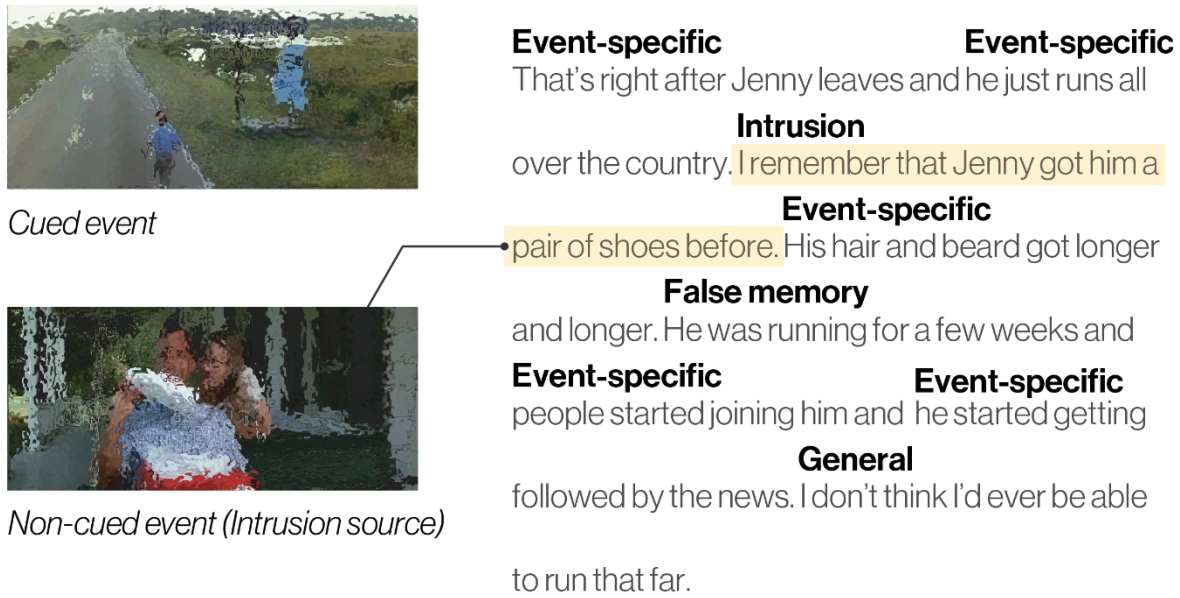


Figure 2. Overview of the detail scoring protocol. Responses on the cued recall task were transcribed and parsed to identify individual details using the protocol described in the Autobiographical Interview scoring protocol (Levine et al., 2002). Details were then categorized as event-specific details, intrusions, false memories, or general details. The event source corresponding to each intrusion was also identified to later quantify its temporal proximity and semantic similarity to the cued event. Images are blurred in this figure for copyright reasons—participants were presented with unblurred images in the study.

Four scorers (BH, MAT, HC, YH) each scored a subset of all the transcripts. To assess interrater reliability, the scorers independently scored the transcripts from 9 participants. These were used to compute single-score intraclass correlations (ICC) assessing the degree of consistency using two-way random effects models (McGraw & Wong, 1996). ICCs were calculated using the *irr* package (Gamer et al., 2012) in R 4.1.0 (R Core Team, 2021). Significant ICCs were observed across all detail types (event-specific details: $ICC(C, 1) = 0.876$, 95% CI [0.855, 0.896], $p < .001$; intrusions: $ICC(C, 1) = 0.875$, 95% CI [0.853, 0.894], $p < .001$; false memories: $ICC(C, 1) = 0.370$, 95% CI [0.311, 0.431], $p < .001$; general details: $ICC(C, 1) =$

0.608, 95% CI [0.557, 0.657], $p < .001$). The low number of false memories and general details provided on average for a given trial may be partially reflected in the lower ICC values observed for these detail types. Summary statistics and ICCs for each detail type are summarized in Table 1.

Table 1. Descriptions, summary statistics, and intraclass correlations for each detail type in the scoring protocol.

Detail Type	Description	M	SD	ICC
Event-specific details	Information (e.g., actions, characters, locations, sounds, dialogue, emotions portrayed, etc.) that is present in the currently cued event	5.01	3.73	.876
Intrusions	Information that is present in the movie, but outside of the currently cued event	2.05	3.74	.875
False memories	Information that is incorrectly remembered from the movie	0.29	0.64	.370
General details	General information that is not specific to the movie, such as general semantic knowledge, repetitions of previously recalled details, and editorializing statements	1.85	2.65	.608

Legend: M = Mean, SD = Standard deviation, ICC = intraclass correlation. Note: Means and standard deviations are calculated across recalls for each cued event.

Statistical analyses

We conducted two sets of statistical analyses to assess whether temporal proximity and life period congruency affected both (1) the number of details recalled for each of the four detail types, and (2) the source of intrusions provided.

Number of details

To assess whether the number of details recalled differed across conditions, we used 2-level multilevel generalized Poisson models nesting individual trials within participants. Poisson

models were used to account for the count-based nature of the number of details recalled (Bolker et al., 2009). According to Barr et al. (2013), we fit models using the maximal random effects structure. For the current study, this entailed estimating (1) fixed effects for temporal proximity, life period congruency, testing session, and their interactions, (2) random slopes for temporal proximity, life period congruency, and their interactions, and (3) random intercepts for each participant. Temporal proximity, life period congruency, and testing session were effect coded (for temporal proximity: Short = -1, Long = 1; for life period congruency: Within = -1, Across = 1; for testing session: Immediate = -1, 24-Hour Delay = 1). Two covariates were also included to control for the duration of each event (log-transformed) and whether participants had previously watched the movie. Separate models were specified for each detail type.

To fit each model, we took a Bayesian approach using the *brms* package (Bürkner, 2017, 2018) in R 4.1.0 (R Core Team, 2021). Weakly informed priors were used for all model terms—specifically, we used $Normal(0, 10)$ priors for intercept terms, $Normal(0, 1)$ priors for beta terms, $half-Cauchy(0, 1)$ for standard deviation terms, and $LKJ(1)$ priors for correlations between random effects. Posterior distributions were obtained using Markov Chain Monte Carlo sampling with chains of 10000 iterations each—2000 iterations in each chain were discarded for warmup. For all parameters, $\hat{R}$ values were equal to 1, suggesting convergence on the posterior distributions. R^2 values, as adapted for Bayesian models, were also calculated to assess overall model fit (Gelman et al., 2019).

For each effect, we report the beta estimates, 95% highest density interval (HDI), and probability of direction (pd). The 95% HDI provides a range of values where the estimate lies with a 95% probability. The pd represents the proportion of the posterior distribution that lies in the same direction (i.e., below or above 0) as the estimate, providing a metric for the existence of

an effect (Makowski, Ben-Shachar, Chen, et al., 2019). To quantify the degree of support for a given effect, we report its Savage-Dickey Bayes Factor, which compares the posterior distribution of a given effect against the point null (i.e., 0) (Wagenmakers et al., 2010). Bayes Factors (BF) were interpreted so that values greater than 1 represented anecdotal evidence in favour of a given hypothesis, values greater than 3 represented substantial evidences in favour of a given hypothesis, values greater than 10 represented strong evidence in favour of a given hypothesis, values greater than 30 represented very strong evidence in favour of a given hypothesis, and values greater than 100 represented decisive evidence in favour of a given hypothesis (Jeffreys, 1961; Wetzels et al., 2011). 95% HDIs, *pds*, and Savage-Dickey Bayes Factors were computed using the *bayestestR* package (Makowski, Ben-Shachar, & Lüdtke, 2019).

Bayes R^2 values and fixed effect estimates, along with their corresponding 95% HDIs, *pds*, and Savage-Dickey BFs, for models predicting each detail type are summarized in Table S1. Only interactions where the model suggests there is some evidence of an effect (i.e., $BF > 1$) are described. Simple effects tests were used to probe these interactions using the *emmeans* package (Lenth et al., 2021).

Source of intrusions: Temporal proximity

To test whether the temporal proximity of intrusion events differed across conditions, we calculated the probability of providing an intrusion as a function of its event lag, or distance in number of events, from the cued event. Positive event lags indicate an intrusion from an event taking place after the cued event while negative event lags indicate an intrusion from an event taking place before the cued event.

To quantify the shape of these functions, we fit a power function to each participant's intrusion probability function, similar to the approach used to quantify temporal contiguity for the free recall of word lists (Howard et al., 2007; Sadeh et al., 2015). This was done separately for positive (1 to 5) and negative (-1 to -5) event lags. The power function took the form of $P(lag) = A|lag|^b$, with the magnitude of the exponent B representing the steepness of the probability function. We added a constant of 1 to each probability to account for probabilities of zero for a given event lag. In addition, we calculated the average intrusion probability for adjacent (event lags with absolute value of 1) and remote (event lags with absolute value between 2 and 5) event lag positions (Kahana et al., 2002).

We then used a Bayesian 2-level multilevel model to compare both the steepness of these functions and the intrusion probabilities for adjacent and remote lags across levels of temporal proximity, life period congruency, and testing session. A random intercept was estimated for each participant. A zero-one-inflated beta distribution was used to best model the characteristics of intrusion probabilities (Liu & Kong, 2015; Ospina & Ferrari, 2012). Separate models were run for both positive and negative lag components, and for adjacent and remote lags.

Source of intrusions: Semantic similarity

In addition to the temporal proximity of intrusion events, we wanted to assess whether semantic similarity to the cued event affected the source of intrusion events different across conditions. To quantify semantic similarity, we first transcribed the described video for the movie, an additional audio narration that describes visual elements (e.g., objects, actions, locations, portrayed emotions, etc.) for those who are visually impaired, and segmented this to align with each event. We then used each event description as an input to Universal Sentence Encoder v4 (Cer et al., 2018), which encodes multiword text input into a 512-dimensional vector. This high-dimensional

vector acts as a representation of the initial text input, with the cosine similarity between a pair of vectors capturing the semantic similarity between their respective inputs. To verify this metric of semantic similarity, we used a Bayesian zero-one-inflated beta regression to compare the average pairwise cosine similarity of events that are within a life period boundary to across a life period boundary—as would be expected, we found decisive evidence for greater semantic similarity between events that are within a life period boundary compared to those that are across a life period boundary ($b = 0.20$, 95% HDI $[0.16, 0.24]$, $pd = 100\%$, $BF = 2.36 \times 10^9$).

We then calculated the probability of providing an intrusion as a function of the semantic similarity between an event and the cued event, as captured by the cosine similarity between their respective Universal Sentence Encoder vectors. To quantify the shape of these functions, we took a similar approach to the method described above and fit a linear regression to each participant's intrusion probability function, with the magnitude of the slope representing the steepness of the probability function. We then used a Bayesian 2-level multilevel generalized zero-one-inflated beta model to compare the steepness of these functions across levels of temporal proximity, life period congruency, and testing session. A random intercept was estimated for each participant.

Specifications for all models characterizing the source of intrusions were the same as those previously mentioned to compare detail counts (e.g., contrast codes, weakly informed priors, etc.). For all parameters, $\hat{R}$ values were equal to 1, suggesting convergence on the posterior distributions. Bayes R^2 values and fixed effect estimates, along with their corresponding 95% HDIs, pds , and Savage-Dickey BFs, are reported in Table S2 for models characterizing the intrusion probabilities as a function of event lags, and in Table S3 for the model characterizing the intrusion probabilities as a function of USE similarity. Only

interactions where the model suggests there is some evidence of an effect (i.e., $BF > 1$) are described. Simple effects tests were used to probe these interactions using the *emmeans* package (Lenth et al., 2021).

Results

Number of details

Event-specific details

Starting with event-specific details, we found decisive evidence for a main effect of life period congruency, with participants providing fewer event-specific details when the preceding tested event was across a life period boundary compared to when the preceding event was within a life period boundary ($b = -0.09$, 95% HDI $[-0.12, -0.07]$, $pd = 100\%$, $BF = 2.11 \times 10^4$) (Figure 3A).

We found no evidence for a main effect of temporal proximity ($b = 0.03$, 95% HDI $[0.00, 0.06]$, $pd = 98.93\%$, $BF = 0.189$) or testing session ($b = -0.08$, 95% HDI $[-0.18, 0.02]$, $pd = 93.18\%$, $BF = 0.161$). We also found no evidence for any interactions (all $pds < 97.5\%$, all $BFs < 1$).

Intrusions

In contrast, we found no evidence for a main effect of life period congruency ($b = 0.07$, 95% HDI $[0.01, 0.13]$, $pd = 98.49\%$, $BF = 0.371$) on the number of intrusions recalled (Figure 3B).

However, we found decisive evidence for a main effect of temporal proximity, with participants providing more intrusions when the preceding event was a long lag away compared to when the preceding event was a short lag away ($b = 0.43$, 95% HDI $[0.35, 0.51]$, $pd = 100\%$, $BF = 3.28 \times 10^8$). We found weak evidence for an interaction between temporal proximity and life period congruency ($b = 0.07$, 95% HDI $[0.02, 0.12]$, $pd = 99.67\%$, $BF = 1.09$). When preceded

by events that were a long lag away, participants provided more intrusions when the preceding event was also across a life period boundary than within a life period boundary ($b = 0.275$, 95% HDI [0.158, 0.386])—this difference for life period congruency was not seen when the preceding event was a short lag away ($b = -0.002$, 95% HDI [-0.193, 0.189]). We found no evidence for a main effect of testing session ($b = 0.09$, 95% HDI [-0.09, 0.27], $pd = 84.93\%$, $BF = 0.154$) or for any other interactions (all $pds < 97.5\%$, all $BFs < 1$).

False memories

For false memories, we found weak evidence for a main effect of life period congruency, with participants providing more false memories when the preceding event was within a life period boundary compared to when the preceding event was across a life period boundary ($b = -0.14$, 95% HDI [-0.25, -0.03], $pd = 99.34\%$, $BF = 1.20$) (Figure 3C). We found no evidence for a main effect of temporal proximity ($b = 0.13$, 95% HDI [0.01, 0.25], $pd = 98.43\%$, $BF = 0.664$) or testing session ($b = 0.06$, 95% HDI [-0.16, 0.27], $pd = 70.68\%$, $BF = 0.124$). However, we found weak evidence for an interaction between temporal proximity and testing session ($b = 0.13$, 95% HDI [0.03, 0.24], $pd = 99.24\%$, $BF = 1.15$). Participants who were tested 24 hours after first watching the movie provided more false memories when the preceding event was a long lag away than when the preceding event was a short lag away ($b = 0.520$, 95% HDI [0.217, 0.835])—this difference in false memories was not seen for participants that were tested immediately after watching the movie ($b = -0.003$, 95% HDI [-0.315, 0.307]). We found no evidence for any other interactions (all $pds < 97.5\%$, all $BFs < 1$).

General details

For general details, we found no evidence for a main effect of temporal proximity ($b = 0.04$, 95% HDI [-0.02, 0.09], $pd = 90.49\%$, $BF = 0.067$), life period congruency ($b = -0.03$, 95% HDI [-

0.08, 0.01], $pd = 93.58\%$, $BF = 0.070$), or testing session ($b = 0.20$, 95% HDI [0.01, 0.37], $pd = 98.24\%$, $BF = 0.934$) (Figure 3D). We also found no evidence for any interactions (all $pds < 97.5\%$, all $BFs < 1$).

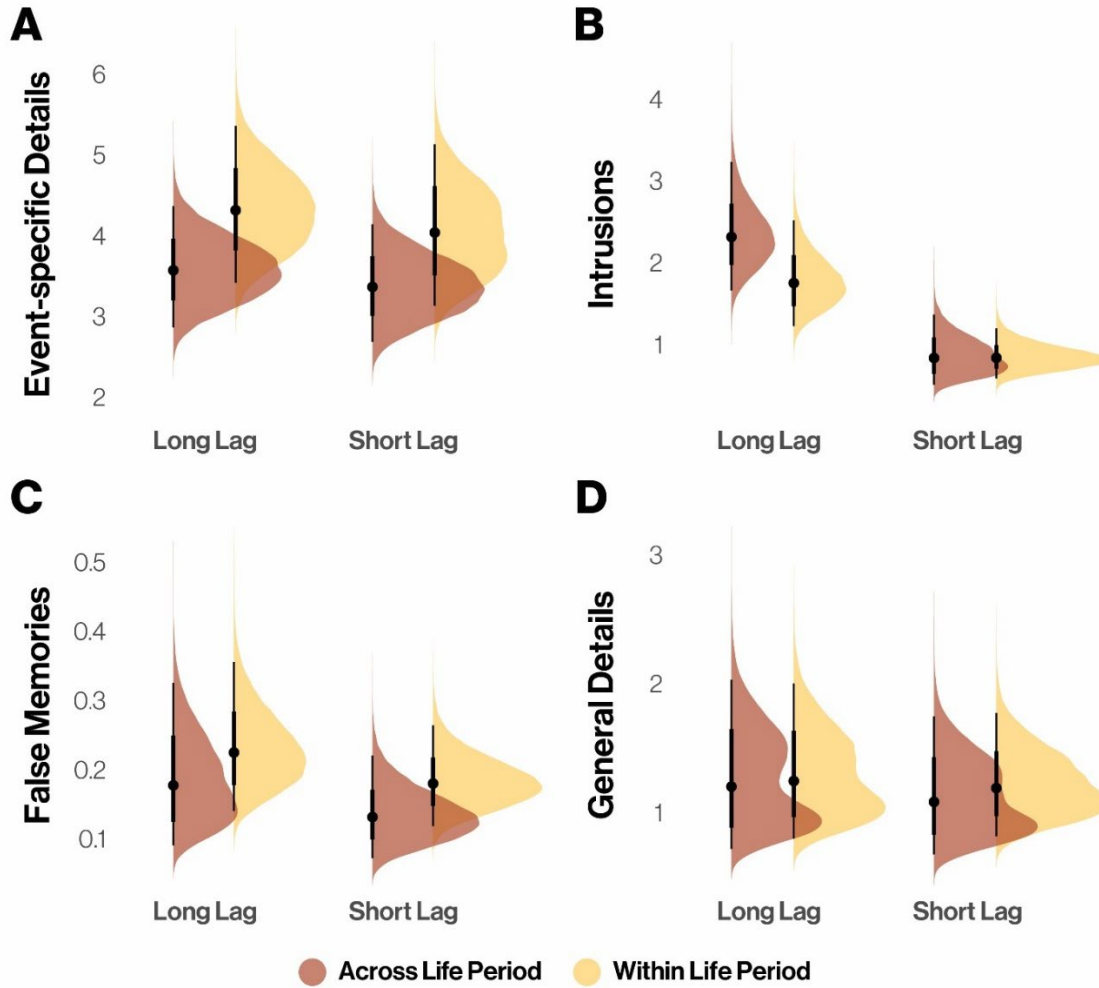


Figure 3. (A) Event-specific details, (B) intrusions, (C) false memories, and (D) general details across different levels of temporal proximity (preceding event was a long or short lag away) and life period congruency (preceding event was across or within life period). Participants provided more event-specific details and false memories when the preceding event was within the same life period. Additionally, participants provided more intrusions when the preceding event was a long lag away, with this effect exacerbated if the preceding event was also across a life period boundary. Density plots depict conditional posterior distributions, with point estimates representing conditional posterior medians. Error bars represent 66% (inner) and 95% (outer) credible intervals. Conditional posterior distributions were estimated across testing sessions (immediate or 24-hour delay) for visualization purposes.

*Source of intrusions*Temporal proximity

Qualitatively, we observed a lag recency effect across all conditions, as evident by the peaked shape of each intrusion probability function, indicating that participants were more likely to provide an intrusion from events that were close by in time to the cued event than events that were further away (Figure 4). However, we see an asymmetry in this pattern, with intrusions being more likely to come from a future event in the movie when the event that had been cued on the prior trial was a short lag away, but more likely to come from a previous event in the movie when the event that had been cued on the prior trial was a long lag away. Within each temporal proximity condition, life period congruency influences the steepness of each intrusion probability curve. When the preceding event was a short lag away, there is a steeper lag recency effect in the forward direction when the event also crosses into a different life period. However, when the preceding event was a long lag away, there is a steeper lag recency effect in the reverse direction when the event was within the same life period.

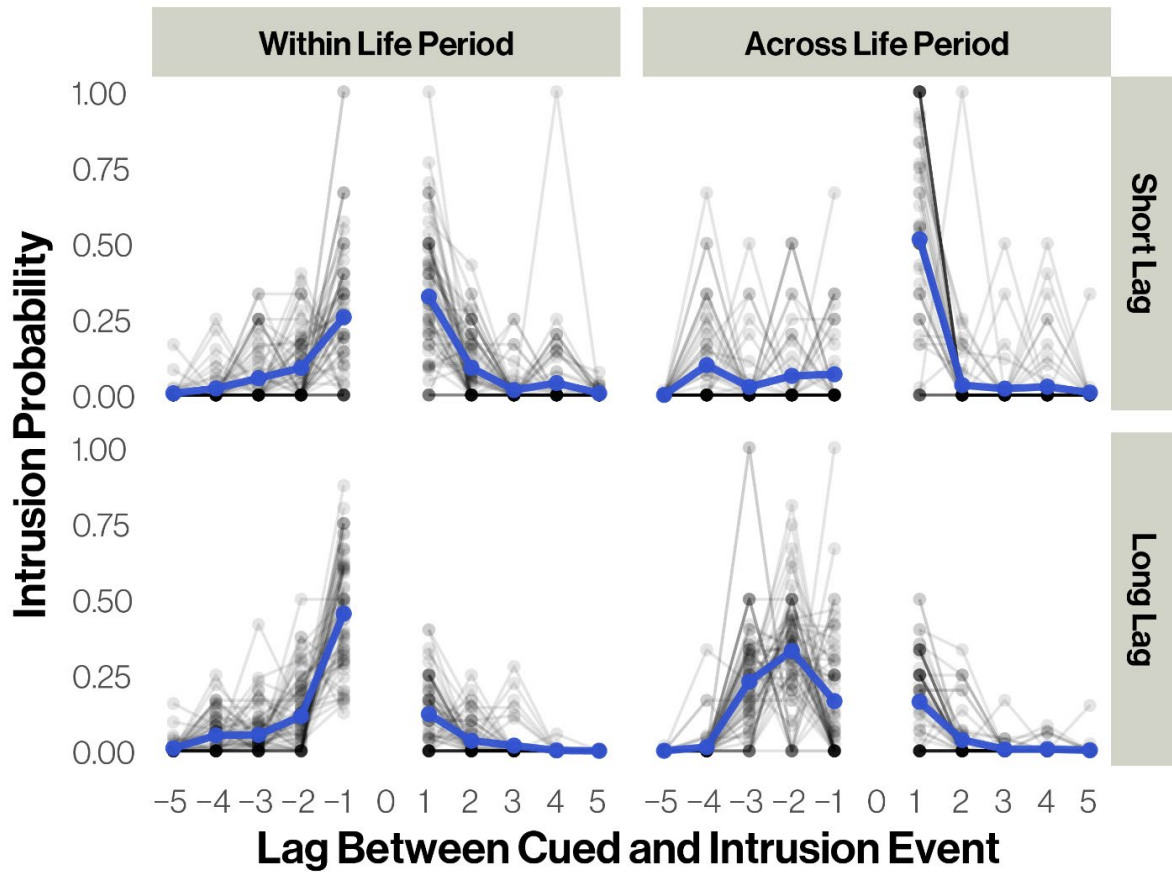


Figure 4. Intrusion probabilities as a function of temporal proximity of the cued event to the intrusion event, as quantified by the lag between the intrusion event and the cued event. Participants were more likely to produce intrusions from future events when the previously cued event was a short lag away—conversely, they were more likely to produce intrusions from events prior events when the previously cued event was a long lag away. Mean intrusion probability functions are denoted by blue points and lines. Error bars denote one standard error. Individual intrusion probability functions are denoted as grey points and lines. Only intrusions from ± 5 event lags are plotted in this figure. Results were collapsed across testing sessions (immediate or 24-hour delay) for visualization purposes.

For positive event lags between the cued event and the intrusion event (i.e., intrusions from events that occurred after the cued event), we found decisive evidence for a main effect of temporal proximity ($b = 0.06$, 95% HDI [0.05, 0.07], $pd = 100\%$, $BF = 6.91 \times 10^8$) and strong evidence for a main effect of life period congruency ($b = -0.02$, 95% HDI [-0.03, -0.01], $pd = 99.99\%$, $BF = 10.26$) on the steepness of the intrusion probability function. Participants had a

steeper curve for positive event lags when the preceding event was either a short lag away compared to a long lag away, or across a life period boundary compared to within a life period boundary. There was no evidence for a main effect of testing session ($b = -0.02$, 95% HDI $[-0.03, 0.00]$, $pd = 97.34\%$, $BF = 0.055$). However, there was strong evidence for the interaction between temporal proximity and testing session ($b = 0.02$, 95% HDI $[0.01, 0.03]$, $pd = 100\%$, $BF = 16.40$). Although all participants showed a steeper curve when the preceding event was a short lag away compared to a long lag away, participants with an immediate testing session showed a larger difference in steepness between the two temporal proximity conditions ($b = 0.165$, 95% HDI $[0.135, 0.196]$) than participants with a 24-hour delay testing session ($b = 0.0716$, 95% HDI $[0.0396, 0.104]$). We found no evidence for any other interactions (all $pds < 97.5\%$ except for the interaction between temporal proximity and life period congruency, with a $pd = 98.96\%$, all $BFs < 1$).

For negative event lags between the cued event and the intrusion event (i.e., intrusions from events that occurred before the cued event), we found decisive evidence for a main effect of temporal proximity ($b = -0.04$, 95% HDI $[-0.05, -0.03]$, $pd = 100\%$, $BF = 9.94 \times 10^5$) and life period congruency ($b = 0.06$, 95% HDI $[0.05, 0.07]$, $pd = 100\%$, $BF = 1.80 \times 10^9$) on the steepness of the intrusion probability function. Participants demonstrated a steeper curve for negative event lags when the preceding event that had been tested was either a long lag away compared to a short lag away, or within a life period boundary compared to across a life period boundary. There was no evidence for a main effect of testing session ($b = -0.02$, 95% HDI $[-0.03, 0.00]$, $pd = 99.60\%$, $BF = 0.0204$). We also found no evidence for any interactions (all $pds < 97.5\%$, all $BFs < 1$).

An analysis of the intrusion probabilities from events that were adjacent (± 1) or remote ($\pm 2-5$) to the cued event was conducted to further characterize the source of intrusion details (Table 2; Table S2). Notably, we found evidence for an interaction between temporal proximity and life period congruency on the intrusion probability from negative event lags, but not positive lags. Specifically, we found weak evidence for this interaction for negative adjacent event lags ($b = -0.19$, 95% HDI $[-0.33, -0.05]$, $pd = 99.55\%$, $BF = 2.50$) and strong evidence for this interaction for negative remote event lags ($b = -0.04$, 95% HDI $[0.06, 0.23]$, $pd = 99.96\%$, $BF = 10.71$). Participants were more likely to provide an intrusion from negative adjacent lags when the preceding cued event was both a long lag away and within a life period boundary ($b = 0.659$, 95% HDI $[0.383, 0.928]$)—however, this was not seen when the preceding cued event was across a life period boundary ($b = -0.123$, 95% HDI $[-0.615, 0.371]$). Conversely, participants were more likely to provide an intrusion from negative remote lags when the preceding cued event was both a long lag away and across a life period boundary ($b = 0.625$, 95% HDI $[0.399, 0.860]$)—however, this was not seen when the preceding cued event was within a life period boundary ($b = 0.0517$, 95% HDI $[-0.181, 0.281]$).

Table 2. Conditional posterior medians (and 95% HDIs) for intrusion probabilities from events that were adjacent (± 1) or remote ($\pm 2-5$) to the cued event.

		Positive lags between cued and intrusion events		Negative lags between cued and intrusion events	
		Adjacent (+1)	Remote (2 – 5)	Adjacent (-1)	Remote (-2 – -5)
Within life period	Short lag	0.343 [0.298, 0.391]	0.055 [0.045, 0.066]	0.299 [0.254, 0.347]	0.058 [0.049, 0.070]
	Long lag	0.183 [0.142, 0.225]	0.027 [0.019, 0.035]	0.451 [0.406, 0.496]	0.061 [0.052, 0.071]
Across life periods	Short lag	0.518 [0.460, 0.576]	0.067 [0.052, 0.085]	0.263 [0.181, 0.347]	0.085 [0.069, 0.102]
	Long lag	0.246 [0.199, 0.298]	0.040 [0.028, 0.054]	0.240 [0.193, 0.288]	0.148 [0.133, 0.163]

Semantic similarity

Qualitatively, we find a positive slope in the intrusion probabilities as a function of semantic similarity across all conditions, indicating that participants were more likely to provide an intrusion from events that were semantically similar to the cued event (Figure 5). This corroborates with our analysis finding decisive evidence for a positive intercept, indicating a positive relationship between USE similarity and intrusion probability ($b = 0.07$, 95% HDI [0.07, 0.08], $pd = 100\%$, $BF = 3.60 \times 10^{38}$). However, we found no evidence for any main effects or interactions on this relationship (all $pds < 97.5\%$ except for the interaction between life period congruency and testing session, with a $pd = 98.88\%$, all $BFs < 1$).

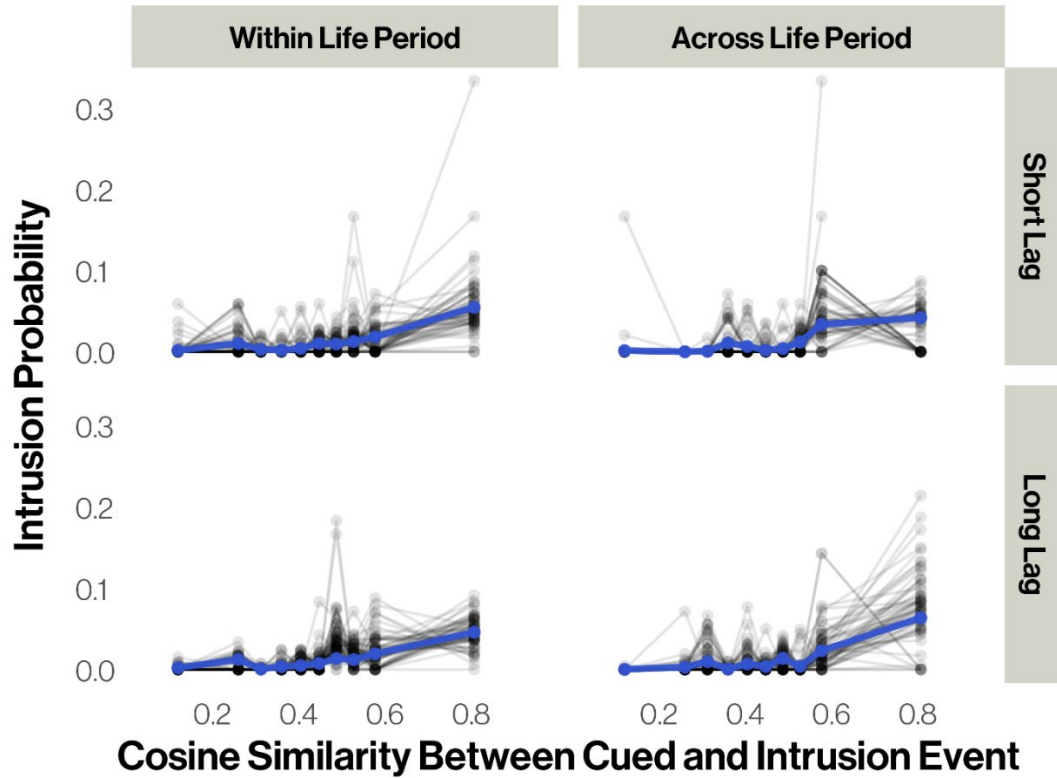


Figure 5. Intrusion probabilities as a function of semantic similarity between the cued event and the intrusion event, as quantified by the cosine similarity between Universal Sentence Encoder (USE) embeddings for the intrusion event and the cued event. Participants were more likely to produce intrusions from events that were semantically similar to the cued event. For visualization purposes, USE similarities were grouped into 10 bins, with each bin being defined so that the number of items in each bin is relatively equal. Each point represents the average intrusion probability for a given bin. Mean intrusion probability functions are denoted by blue points and lines. Error bars denote one standard error. Individual intrusion probability functions are denoted as grey points and lines.

Discussion

The results in the present study suggest that the detailed recall for a naturalistic event is affected by its similarity to the event that had been recalled immediately prior. Here, we manipulated the trial order during a cued recall task to vary the temporal proximity and life period congruency of adjacently tested events from the movie *Forrest Gump*. We found that

participants provided more event-specific details and false memories when the preceding tested event was within, compared to across, life periods. Participants also tended to provide more intrusions when the preceding tested event was a long, compared to short, temporal distance away—this increase in intrusions was exacerbated if the preceding event was also across a life period. Moreover, the source of these intrusions varied depending on both the temporal proximity and the life period congruency of the preceding cued event.

The observed differences in the number event-specific details and intrusions recalled supports our initial hypothesis of improved episodic richness for consecutively tested events with high similarity. Specifically, we found that participants produced more event-specific details when the preceding cued event was within the same life period and fewer intrusions when the preceding cued event was a short temporal distance away, suggesting that the shared context between the two may have facilitated better recollection. By demonstrating that memory strength can be affected by directly manipulating the similarity between adjacent events at recall, we extend previous work showing associations between the degree of clustering at recall and overall memory performance (Diamond & Levine, 2020; Healey et al., 2014; Healey & Uitvlugt, 2019; Sadeh et al., 2015; Sederberg et al., 2010). We also found that participants provided more false memories when the preceding tested event shared a life period with the cued event, which may reflect reinstatement of a schematic event representation leading to misattribution of more typical schematic information to the movie itself (Bartlett, 1932; Gilboa & Marlatte, 2017; van Kesteren et al., 2012). This provides evidence that the reinstatement of the contextual associations between events at encoding are not a by-product of episodic recall, but rather may be acting as an initial scaffold to support more detailed recollection.

Contrary to our initial predictions, close temporal proximity between adjacently cued events did not produce an increase in the number of event-specific details recalled. However, this may reflect the high degree of semantic similarity between events in the movie given its strong narrative structure (Bellana et al., 2022). This may be analogous to previous studies comparing the free recall of word lists comprised of semantically unrelated items to semantically related items, showing that the long-standing semantic associations between items formed across one's lifetime experience can supersede the temporal associations between items formed at encoding during retrieval (Healey & Uitvlugt, 2019; Polyn et al., 2011). Similar results have also been observed for recall of autobiographical events which can be organized within a broader personal narrative, with no benefit in episodic recall being found for consecutively cued personal events that were close in time (Hong et al., 2024). In situations where there is minimal semantic structure across events, such as when watching a series of unrelated television episodes or short films (Lee & Chen, 2022), the temporal associations between events may play a more central role in facilitating rich episodic recall.

Turning to intrusions, we found that manipulating the degree of similarity between adjacently tested events modulated the source of intrusions from in a way that extends predictions from retrieved-context models. Specifically, participants were more likely to produce intrusions from the movie after the cued event when the preceding tested event was a short lag away. This effect was stronger if the preceding event also crossed over a life period, suggesting that participants were unlikely to produce details from events in a different life period. This pattern of results mirrors the canonical forward bias found in word list free recall tasks, where there is a higher likelihood to transition to items that are forwards in time rather than backwards (Healey et al., 2018; Howard & Kahana, 2002; Kahana, 1996).

Interestingly, the observed forward bias in intrusions flips when adjacently tested events were far away in time. Here, participants tended to draw details from events that occurred before the cued event in the movie when the preceding tested event was a long lag away, with this backwards bias being stronger when the preceding event was also from the same life period. Given the narrative structure of the movie, which encourages recall in a forward direction from beginning to end (Lee et al., 2020), these intrusions may reflect the participant having to think back to fill in the details from the narrative gap resulting from consecutive recall of events that are far away in time. This also helps explain the differences in the strength of the temporal contiguity effect observed depending on the life period congruency of the previously cued event. If the event tested on the trial immediately prior was from the same life period, participants may need to draw only on information from a nearby event to facilitate detailed recall for the cued event. However, if the preceding tested event is from a different life period, the temporal contiguity effect is weaker, suggesting that participants may need require additional support from information spread out across the gap between adjacently cued events. Future work can further probe the trajectory of details within the recall of a given event to investigate whether intrusions are provided to support further episodic recall (Diamond & Levine, 2020; Heusser et al., 2021). For example, if participants are primarily providing intrusions prior to event-specific details, this suggests that participants are using intrusions to provide temporal structure to help facilitate detailed recall for the cued trial; however, if participants are primarily providing event-specific details prior to intrusions, this suggests that participants are using intrusions to provide related, but superfluous, information to their recall.

Further, the observed differences in the source of intrusions produced show that the life periods may be modulating similarity between events beyond temporal proximity. Here, we

found that participants tended to draw on information from events with a shared life narrative when making intrusions during recall. For example, participants will be more likely to produce intrusions for adjacent events in the forward direction when the preceding cued event was both a short lag away and across a life period—further, few intrusions came from events that took place before the cued event. This pattern suggests that participants were unlikely to draw from events coming from a different life period. These findings are in line with evidence demonstrating how event boundaries can produce recall transitions that are driven more by event similarity rather than temporal similarity (Heusser et al., 2018; Lohnas et al., 2023; Polyn et al., 2009a, 2009b), with recent work pointing towards a potential mechanism of a resetting of temporal context at event boundaries (Pu et al., 2022).

Moreover, we found that participants were more likely to provide intrusions from events that had high semantic similarity to the cued event. This is consistent with previous work showing how semantic associations are used in conjunction with temporal associations during memory retrieval (Howard et al., 2007; Polyn et al., 2009a; Sederberg et al., 2010). Notably, the strength of this effect was the same regardless of the similarity between the cued and preceding event in terms of the temporal lag and life period congruency. This may be driven by the strong semantic similarity across events in the movie—for example, semantic associations between items have been shown to overpower temporal associations between items in word lists made up of semantically related items (Healey & Uitvlugt, 2019; Polyn et al., 2011). Using this same paradigm with a movie with weaker semantic associations between events may reveal a stronger reliance on semantic associations when temporal associations are less readily available. Furthermore, future work can investigate whether other types of associations in *Forrest Gump*, such as emotion (Labs et al., 2015; Lohnas & Howard, 2024; Long et al., 2015; Talmi et al.,

2019; Tomita et al., 2021) or more overarching narrative structures (Baldassano et al., 2018; Milivojevic et al., 2016), are being used to guide recall.

Interestingly, we did not observe strong evidence for any moderating effects of interest from varying the timing of testing sessions after having watched the movie. Extensive work has shown a crucial role of sleep in making memories for recent events more resistant to forgetting (Diekelmann & Born, 2010; Stickgold, 2005). In this process, the shared features across these different events may be extracted and integrated into an existing schematic representation (Lewis & Durrant, 2011). Here, we predicted that this abstraction of information with sleep across testing sessions could lead participants to better pinpoint how individual events fit in relation to their overarching life period, thus magnifying any potential effects from life period congruency on recall (Cohn-Sheehy et al., 2021; Liu & Ranganath, 2021). However, we did not observe any notable differences in recall towards this prediction between participants tested immediately, compared to 24 hours, after watching the movie. One potential explanation for this may be the popularity of the movie *Forrest Gump*, which may result in participants having some existing schematic information. Future work could address this by using this task paradigm with other movies sharing a similar narrative structure—for example, movies like *Boyhood* (R. Linklater, IFC Films/Universal Pictures, 2014) and *Moonlight* (B. Jenkins, A24, 2016) both follow the progression of their respective protagonists throughout various periods of their lives.

In the current study, we investigated the influence of temporal proximity and schema on memory for events within a broader, hierarchically structured narrative. Here, we find that the detailed recall for a given event in the movie *Forrest Gump* was affected by its similarity to the event that was recalled just prior, whether by being close in time or in the same life period. These

results demonstrate how the temporal and schematic organization of memory thread together and shape how we recount events from our past.

References

- Averell, L., Prince, M., & Heathcote, A. (2016). Fundamental causes of systematic and random variability in recognition memory. *Journal of Memory and Language*, 88, 51–69.
<https://doi.org/10.1016/j.jml.2015.12.010>
- Baldassano, C., Chen, J., Zadbood, A., Pillow, J. W., Hasson, U., & Norman, K. A. (2017). Discovering event structure in continuous narrative perception and memory. *Neuron*, 95(3). <https://doi.org/10.1016/j.neuron.2017.06.041>
- Baldassano, C., Hasson, U., & Norman, K. A. (2018). Representation of real-world event schemas during narrative perception. *The Journal of Neuroscience*, 38(45).
<https://doi.org/10.1101/252718>
- Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, 68(3). <https://doi.org/10.1016/j.jml.2012.11.001>
- Bartlett, F. C. (1932). *Remembering: A Study in Experimental and Social Psychology*. Cambridge University Press.
- Bellana, B., Mahabal, A., & Honey, C. J. (2022). Narrative thinking lingers in spontaneous thought. *Nature Communications*, 13(1). <https://doi.org/10.1038/s41467-022-32113-6>
- Ben-Yakov, A., & Henson, R. N. (2018). The hippocampal film-editor: Sensitivity and specificity to event boundaries in continuous experience. *Journal of Neuroscience*, 38(47). <https://doi.org/10.1101/273409>
- Bolker, B. M., Brooks, M. E., Clark, C. J., Geange, S. W., Poulsen, J. R., Stevens, M. H. H., & White, J.-S. S. (2009). Generalized linear mixed models: A practical guide for ecology

- and evolution. *Trends in Ecology & Evolution*, 24(3), 127–135.
<https://doi.org/10.1016/j.tree.2008.10.008>
- Brunec, I. K., Ozubko, J. D., Barense, M. D., & Moscovitch, M. (2017). Recollection-dependent memory for event duration in large-scale spatial navigation. *Learning & Memory*, 24(3), 104–114. <https://doi.org/10.1101/lm.044032.116>
- Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, 80, 1–28. <https://doi.org/10.18637/jss.v080.i01>
- Bürkner, P.-C. (2018). Advanced Bayesian multilevel modeling with the R package brms. *The R Journal*, 10(1), 395–411. <https://doi.org/10.32614/RJ-2018-017>
- Cer, D., Yang, Y., Kong, S., Hua, N., Limtiaco, N., John, R. S., Constant, N., Guajardo-Cespedes, M., Yuan, S., Tar, C., Sung, Y.-H., Strope, B., & Kurzweil, R. (2018). *Universal sentence encoder* [Preprint]. arXiv. <http://arxiv.org/abs/1803.11175>
- Clewett, D., DuBrow, S., & Davachi, L. (2019). Transcending time in the brain: How event memories are constructed from experience. *Hippocampus*, 1–22.
<https://doi.org/10.1002/hipo.23074>
- Cohn-Sheehy, B. I., Delarazan, A. I., Crivelli-Decker, J. E., Reagh, Z. M., Mundada, N. S., Yonelinas, A. P., Zacks, J. M., & Ranganath, C. (2021). Narratives bridge the divide between distant events in episodic memory. *Memory & Cognition*.
<https://doi.org/10.3758/s13421-021-01178-x>
- Diamond, N. B., & Levine, B. (2020). Linking detail to temporal structure in naturalistic-event recall. *Psychological Science*, 31(12), 1557–1572.
<https://doi.org/10.1177/0956797620958651>

- Diekelmann, S., & Born, J. (2010). The memory function of sleep. *Nature Reviews Neuroscience*, 11(2). <https://doi.org/10.1038/nrn2762>
- DuBrow, S. (2024). Events and Boundaries. In M. J. Kahana & A. D. Wagner (Eds.), *The Oxford Handbook of Human Memory, Two Volume Pack* (1st ed., pp. 497–519). Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780190917982.013.18>
- DuBrow, S., & Davachi, L. (2013). The influence of context boundaries on memory for the sequential order of events. *Journal of Experimental Psychology: General*, 142(4). <https://doi.org/10.1037/a0034024>
- DuBrow, S., & Davachi, L. (2016). Temporal binding within and across events. *Neurobiology of Learning and Memory*, 134, 107–114. <https://doi.org/10.1016/j.nlm.2016.07.011>
- DuBrow, S., Rouhani, N., Niv, Y., & Norman, K. A. (2017). Does mental context drift or shift? *Current Opinion in Behavioral Sciences*, 17, 141–146. <https://doi.org/10.1016/j.cobeha.2017.08.003>
- Ezzyat, Y., & Davachi, L. (2011). What constitutes an episode in episodic memory? *Psychological Science*, 22(2). <https://doi.org/10.1177/0956797610393742>
- Ezzyat, Y., & Davachi, L. (2014). Similarity breeds proximity: Pattern similarity within and across contexts is related to later mnemonic judgments of temporal proximity. *Neuron*, 81(5). <https://doi.org/10.1016/j.neuron.2014.01.042>
- Faber, M., & Gennari, S. P. (2015). In search of lost time: Reconstructing the unfolding of events from memory. *Cognition*, 143, 193–202. <https://doi.org/10.1016/j.cognition.2015.06.014>
- Gamer, M., Lemon, J., & Singh, P. (2012). *irr: Various Coefficients of Interrater Reliability and Agreement* (Version Version 0.84.1) [R Package]. <https://cran.r-project.org/web/packages/irr/irr.pdf>

- Gelman, A., Goodrich, B., Gabry, J., & Vehtari, A. (2019). R-squared for Bayesian regression models. *The American Statistician*, 73(3), 307–309.
<https://doi.org/10.1080/00031305.2018.1549100>
- Gilboa, A., & Marlatte, H. (2017). Neurobiology of schemas and schema-mediated memory. *Trends in Cognitive Sciences*, 21(8). <https://doi.org/10.1016/j.tics.2017.04.013>
- Hanke, M., Adelhöfer, N., Kottke, D., Iacovella, V., Sengupta, A., Kaule, F. R., Nigbur, R., Waite, A. Q., Baumgartner, F., & Stadler, J. (2016). A studyforrest extension, simultaneous fMRI and eye gaze recordings during prolonged natural stimulation. *Scientific Data*, 3(160092). <https://doi.org/10.1038/sdata.2016.92>
- Healey, M. K., Crutchley, P., & Kahana, M. J. (2014). Individual differences in memory search and their relation to intelligence. *Journal of Experimental Psychology: General*, 143(4). <https://doi.org/10.1037/a0036306>
- Healey, M. K., Long, N. M., & Kahana, M. J. (2018). Contiguity in episodic memory. *Psychonomic Bulletin & Review*, 1–22. <https://doi.org/10.3758/s13423-018-1537-3>
- Healey, M. K., & Uitvlugt, M. G. (2019). The role of control processes in temporal and semantic contiguity. *Memory and Cognition*, 1–19. <https://doi.org/10.3758/s13421-019-00895-8>
- Heusser, A. C., Ezzyat, Y., Shiff, I., & Davachi, L. (2018). Perceptual boundaries cause mnemonic trade-offs between local boundary processing and across-trial associative binding. *Journal of Experimental Psychology: Learning Memory and Cognition*, 44(7). <https://doi.org/10.1037/xlm0000503>
- Heusser, A. C., Fitzpatrick, P. C., & Manning, J. R. (2021). Geometric models reveal behavioural and neural signatures of transforming experiences into memories. *Nature Human Behaviour*, 1–15. <https://doi.org/10.1038/s41562-021-01051-6>

- Hong, B., Tran, M. A., Cheng, H., Arenas Rodriguez, B., Li, K. E., & Barense, M. D. (2024). The influence of event similarity on the detailed recall of autobiographical memories. *Memory*, 1–13. <https://doi.org/10.1080/09658211.2024.2406307>
- Howard, M. W., Addis, K. M., Jing, B., & Kahana, M. J. (2007). Semantic structure and episodic memory. In T. K. Landauer, D. S. McNamara, S. J. Dennis, & W. Kintsch (Eds.), *Handbook of Latent Semantic Analysis* (pp. 121–141). Psychology Press.
- Howard, M. W., & Kahana, M. J. (2002). A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46(3). <https://doi.org/10.1006/jmps.2001.1388>
- Howard, M. W., Shankar, K. H., Aue, W. R., & Criss, A. H. (2015). A distributed representation of internal time. *Psychological Review*, 122(1). <https://doi.org/10.1037/a0037840>
- Jeffreys, H. (1961). *Theory of Probability* (3rd ed.). Oxford University Press.
- Kahana, M. J. (1996). Associative retrieval processes in free recall. *Memory & Cognition*, 24(1). <https://doi.org/10.3758/BF03197276>
- Kahana, M. J. (2020). Computational models of memory search. *Annual Review of Psychology*, 71(1). <https://doi.org/10.1146/annurev-psych-010418-103358>
- Kahana, M. J., Howard, M. W., Zaromb, F., & Wingfield, A. (2002). Age dissociates recency and lag recency effects in free recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 530–540.
- Kurby, C. A., & Zacks, J. M. (2008). Segmentation in the perception and memory of events. *Trends in Cognitive Sciences*, 12(2). <https://doi.org/10.1002/cam4.334>
- Kurby, C. A., & Zacks, J. M. (2011). Age differences in the perception of hierarchical structure in events. *Memory & Cognition*, 39(1). <https://doi.org/10.3758/s13421-010-0027-2>

- Labs, A., Reich, T., Schulenburg, H., Boennen, M., Mareike, G., Golz, M., Hartigs, B., Hoffmann, N., Keil, S., Perlow, M., Peukmann, A. K., Rabe, L. N., von Sobbe, F.-R., & Hanke, M. (2015). Portrayed emotions in the movie “Forrest Gump.” *F1000Research*, 4(92). <https://doi.org/10.12688/f1000research.6230.1>
- Lee, H., Bellana, B., & Chen, J. (2020). What can narratives tell us about the neural bases of human memory? *Current Opinion in Behavioral Sciences*, 32, 111–119. <https://doi.org/10.1016/j.cobeha.2020.02.007>
- Lee, H., & Chen, J. (2022). Predicting memory from the network structure of naturalistic events. *Nature Communications*, 13(1), 4235. <https://doi.org/10.1038/s41467-022-31965-2>
- Lenth, R. V., Buerkner, P., Herve, M., Love, J., Riebl, H., & Singmann, H. (2021). *emmeans: Estimated Marginal Means, aka Least-Squares Means* (Version Version 1.6.1) [R Package]. <https://cran.r-project.org/web/packages/emmeans/emmeans.pdf>
- Levine, B., Svoboda, E., Hay, J. F., Winocur, G., & Moscovitch, M. (2002). Aging and autobiographical memory: Dissociation episodic from semantic retrieval. *Psychology and Aging*, 17(4).
- Lewis, P. A., & Durrant, S. J. (2011). Overlapping memory replay during sleep builds cognitive schemata. *Trends in Cognitive Sciences*, 15(8), 343–351. <https://doi.org/10.1016/j.tics.2011.06.004>
- Liu, F., & Kong, Y. (2015). zoib: An R package for Bayesian inference for beta regression and zero/one inflated beta regression. *The R Journal*, 7(2), 34. <https://doi.org/10.32614/RJ-2015-019>
- Liu, X. L., & Ranganath, C. (2021). Resurrected memories: Sleep-dependent memory consolidation saves memories from competition induced by retrieval practice.

- Psychonomic Bulletin & Review*, 28(6), 2035–2044. <https://doi.org/10.3758/s13423-021-01953-6>
- Lohnas, L. J., Healey, M. K., & Davachi, L. (2023). Neural temporal context reinstatement of event structure during memory recall. *Journal of Experimental Psychology: General*, 152(7), 1840–1872. <https://doi.org/10.1037/xge0001354>
- Lohnas, L. J., & Howard, M. W. (2024). The influence of emotion on temporal context models. *Cognition and Emotion*, 1–29. <https://doi.org/10.1080/02699931.2024.2371075>
- Long, N. M., Danoff, M. S., & Kahana, M. J. (2015). Recall dynamics reveal the retrieval of emotional context. *Psychonomic Bulletin and Review*, 22(5). <https://doi.org/10.3758/s13423-014-0791-2>
- Makowski, D., Ben-Shachar, M., & Lüdecke, D. (2019). bayestestR: Describing effects and their uncertainty, existence and significance within the Bayesian framework. *Journal of Open Source Software*, 4(40), 1541. <https://doi.org/10.21105/joss.01541>
- Makowski, D., Ben-Shachar, M. S., Chen, S. H. A., & Lüdecke, D. (2019). Indices of effect existence and significance in the Bayesian framework. *Frontiers in Psychology*, 10, 2767. <https://doi.org/10.3389/fpsyg.2019.02767>
- McGraw, K. O., & Wong, S. P. (1996). Forming inferences about some intraclass correlation coefficients. *Psychological Methods*, 1(1), 30–46.
- Milivojevic, B., Varadinov, M., Vicente Grabovetsky, A., Collin, S. H., & Doeller, C. F. (2016). Coding of event nodes and narrative context in the hippocampus. *The Journal of Neuroscience*, 36(49). <https://doi.org/10.1523/JNEUROSCI.2889-15.2016>

- Moreton, B. J., & Ward, G. (2010). Time scale similarity and long-term memory for autobiographical events. *Psychonomic Bulletin and Review*, 17(4).
<https://doi.org/10.3758/PBR.17.4.510>
- Ospina, R., & Ferrari, S. L. P. (2012). A general class of zero-or-one inflated beta regression models. *Computational Statistics & Data Analysis*, 56(6), 1609–1623.
<https://doi.org/10.1016/j.csda.2011.10.005>
- Pathman, T., Decker, L., Parmar, P. K., Adkins, M. C., & Polyn, S. M. (2023). Children's memory "in the wild": Examining the temporal organization of free recall from a week-long camp at a local zoo. *Cognitive Research: Principles and Implications*, 8(1), 6.
<https://doi.org/10.1186/s41235-022-00452-z>
- Polyn, S. M., & Cutler, R. A. (2017). Retrieved-context models of memory search and the neural representation of time. *Current Opinion in Behavioral Sciences*, 17, 203–210.
<https://doi.org/10.1016/j.cobeha.2017.09.007>
- Polyn, S. M., Erlichman, G., & Kahana, M. J. (2011). Semantic cuing and the scale insensitivity of recency and contiguity. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 37(3). <https://doi.org/10.1037/a0022475>
- Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009a). A context maintenance and retrieval model of organizational processes in free recall. *Psychological Review*, 116(1).
<https://doi.org/10.1037/a0014420>
- Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009b). Task context and organization in free recall. *Neuropsychologia*, 47(11).
<https://doi.org/10.1016/j.neuropsychologia.2009.02.013>

- Pu, Y., Kong, X.-Z., Ranganath, C., & Melloni, L. (2022). Event boundaries shape temporal organization of memory by resetting temporal context. *Nature Communications*, 13(1). <https://doi.org/10.1038/s41467-022-28216-9>
- Radvansky, G. A., & Zacks, J. M. (2017). Event boundaries in memory and cognition. *Current Opinion in Behavioral Sciences*, 17, 133–140. <https://doi.org/10.1016/j.cobeha.2017.08.006>
- Sadeh, T., Moran, R., & Goshen-Gottstein, Y. (2015). When items ‘pop into mind’: Variability in temporal-context reinstatement in free-recall. *Psychonomic Bulletin and Review*, 22(3). <https://doi.org/10.3758/s13423-014-0746-7>
- Schwartz, G., Howard, M. W., Jing, B., & Kahana, M. J. (2005). Shadows of the past: Temporal retrieval effects in recognition memory. *Psychological Science*, 16(11). <https://doi.org/10.1111/j.1467-9280.2005.01634.x>
- Sederberg, P. B., Miller, J. F., Howard, M. W., & Kahana, M. J. (2010). The temporal contiguity effect predicts episodic memory performance. *Memory and Cognition*, 38(6). <https://doi.org/10.3758/MC.38.6.689>
- Shankar, K. H., & Howard, M. W. (2012). A scale-invariant internal representation of time. *Neural Computation*, 24(1). https://doi.org/10.1162/NECO_a_00212
- Stern, Y., Katz, R., & Sadeh, T. (2020). Explicit sequence memory in recall of temporally-structured episodes. *Scientific Reports*, 10(1). <https://doi.org/10.1038/s41598-020-59472-8>
- Stickgold, R. (2005). Sleep-dependent memory consolidation. *Nature*, 437(7063), 1272–1278. <https://doi.org/10.1038/nature04286>

- Talmi, D., Lohnas, L. J., & Daw, N. D. (2019). A retrieved context model of the emotional modulation of memory. *Psychological Review*, 126(4), 455–485.
<https://doi.org/10.1037/rev0000132>
- Tomita, T. M., Barense, M. D., & Honey, C. J. (2021). *The similarity structure of real-world memories* [Preprint]. bioRxiv. <https://doi.org/10.1101/2021.01.28.428278>
- Tulving, E. (1972). Episodic and semantic memory. In *Organization of memory* (pp. 381–403). Academic Press.
- Tulving, E. (1985). Memory and consciousness. *Canadian Psychology*, 26(1), 1–12.
<https://doi.org/10.1037/h0080017>
- Tulving, E. (2002). Episodic memory: From mind to brain. *Annual Review of Psychology*, 53(1).
<https://doi.org/10.1146/annurev.psych.53.100901.135114>
- Uitvlugt, M. G., & Healey, M. K. (2018). Temporal proximity links unrelated news events in memory. *Psychological Science*, 30(1), 92–104.
- van Kesteren, M. T. R., Ruiter, D. J., Fernández, G., & Henson, R. N. (2012). How schema and novelty augment memory formation. *Trends in Neurosciences*, 35(4), 211–219.
<https://doi.org/10.1016/j.tins.2012.02.001>
- Wagenmakers, E.-J., Lodewyckx, T., Kuriyal, H., & Grasman, R. (2010). Bayesian hypothesis testing for psychologists: A tutorial on the Savage–Dickey method. *Cognitive Psychology*, 60(3), 158–189. <https://doi.org/10.1016/j.cogpsych.2009.12.001>
- Wetzels, R., Matzke, D., Lee, M. D., Rouder, J. N., Iverson, G. J., & Wagenmakers, E.-J. (2011). Statistical evidence in experimental psychology: An empirical comparison using 855 tests. *Perspectives on Psychological Science*, 6(3), 291–298.
<https://doi.org/10.1177/1745691611406923>

Zacks, J. M., Speer, N. K., Swallow, K. M., Braver, T. S., & Reynolds, J. R. (2007). Event perception: A mind-brain perspective. *Psychological Bulletin*, 133(2).

<https://doi.org/10.1037/0033-2909.133.2.273>

Zacks, J. M., Speer, N. K., Swallow, K. M., & Maley, C. J. (2010). The brain's cutting-room floor: Segmentation of narrative cinema. *Frontiers in Human Neuroscience*, 4(168).

<https://doi.org/10.3389/fnhum.2010.00168>

Zacks, J. M., Tversky, B., & Iyer, G. (2001). Perceiving, remembering and communicating structure in events. *Journal of Experimental Psychology: General*, 130, 29–58.

Appendix

Table S1: Fit statistics and fixed-effects parameters for Bayesian mixed effects models related to detail counts

Detail Type	Bayes R ²	Fixed Effect	Estimate (b)	95% HDI	pd	Savage- Dickey BF
Event-specific details	.443	Intercept	1.45	[1.34, 1.55]	100.00%	3.24×10²⁷
		Temporal Proximity	0.03	[0.00, 0.06]	98.93%	0.189
		Life Period Congruency	-0.09	[-0.12, -0.07]	100.00%	2.11×10⁴
		Testing Session	-0.08	[-0.18, 0.02]	93.18%	0.161
		Log(Event Duration)	0.21	[0.18, 0.23]	100.00%	4.05×10³⁴
		Previously Watched	0.11	[0.00, 0.21]	97.92%	0.453
		Temporal Proximity × Life Period Congruency	0.00	[-0.03, 0.03]	54.58%	0.015
		Temporal Proximity × Testing Session	0.01	[-0.02, 0.03]	65.69%	0.014
		Life Period Congruency × Testing Session	0.01	[-0.01, 0.04]	80.73%	0.018
		Temporal Proximity × Life Period Congruency × Testing Session	0.00	[-0.03, 0.03]	61.31%	0.015
Intrusions	.352	Intercept	0.47	[0.28, 0.65]	100.00%	104.40
		Temporal Proximity	0.43	[0.35, 0.51]	100.00%	3.23×10⁸
		Life Period Congruency	0.07	[0.01, 0.13]	98.49%	0.371
		Testing Session	0.09	[-0.09, 0.27]	84.93%	0.154
		Log(Event Duration)	-0.03	[-0.07, 0.01]	93.07%	0.058
		Previously Watched	0.18	[0.00, 0.36]	97.61%	0.710
		Temporal Proximity × Life Period Congruency	0.07	[0.02, 0.12]	99.67%	1.09

		Temporal Proximity × Testing Session	-0.02	[-0.10, 0.06]	71.95%	0.047
		Life Period Congruency × Testing Session	0.02	[-0.04, 0.08]	76.54%	0.038
		Temporal Proximity × Life Period Congruency × Testing Session	-0.04	[-0.09, 0.01]	94.61%	0.091
False memories	.139	Intercept	-1.55	[-1.76, -1.34]	100.00%	2.25×10¹⁵
		Temporal Proximity	0.13	[0.01, 0.25]	98.43%	0.664
		Life Period Congruency	-0.14	[-0.25, -0.03]	99.32%	1.20
		Testing Session	0.06	[-0.16, 0.27]	70.68%	0.124
		Log(Event Duration)	0.00	[-0.10, 0.10]	50.58%	0.051
		Previously Watched	0.18	[-0.03, 0.38]	95.19%	0.425
		Temporal Proximity × Life Period Congruency	0.02	[-0.09, 0.13]	62.35%	0.060
		Temporal Proximity × Testing Session	0.13	[0.03, 0.24]	99.24%	1.15
		Life Period Congruency × Testing Session	0.01	[-0.09, 0.11]	58.88%	0.052
		Temporal Proximity × Life Period Congruency × Testing Session	0.04	[-0.06, 0.14]	77.71%	0.069
General details	.516	Intercept	0.34	[0.15, 0.53]	99.98%	4.86
		Temporal Proximity	0.04	[-0.02, 0.09]	90.49%	0.067
		Life Period Congruency	-0.03	[-0.08, 0.01]	93.58%	0.070
		Testing Session	0.20	[0.01, 0.37]	98.24%	0.934
		Log(Event Duration)	-0.08	[-0.12, -0.04]	100.00%	36.13
		Previously Watched	0.16	[-0.01, 0.34]	96.25%	0.445
		Temporal Proximity × Life Period Congruency	0.01	[-0.03, 0.05]	73.90%	0.027

Temporal Proximity × Testing Session	0.02	[-0.03, 0.08]	79.48%	0.037
Life Period Congruency × Testing Session	0.03	[-0.01, 0.07]	90.22%	0.047
Temporal Proximity × Life Period Congruency × Testing Session	0.00	[-0.04, 0.04]	56.82%	0.020

Legend: R^2 = coefficient of determination, 95% HDI = 95% highest density interval, pd = probability of direction, Savage-Dickey BF = Savage-Dickey Bayes Factor. Fixed effects with a BF > 1 are bolded.

Table S2: Fit statistics and fixed-effects parameters for Bayesian mixed effects models related to intrusion probabilities based on event lag

Direction	Measure	Bayes R^2	Fixed Effect	Estimate (b)	95% HDI	pd	Savage-Dickey BF
Positive	Power exponent	.488	Intercept	-0.13	[-0.15, -0.12]	100.00%	2.85×10¹⁷
			Temporal Proximity	0.06	[0.05, 0.07]	100.00%	6.48×10⁸
			Life Period Congruency	-0.02	[0.03, -0.01]	99.99%	10.27
			Testing Session	-0.02	[-0.03, 0.00]	97.34%	0.054
			Temporal Proximity × Life Period Congruency	0.01	[0.00, 0.02]	98.96%	0.083
			Temporal Proximity × Testing Session	0.02	[0.01, 0.03]	100.00%	16.71
			Life Period Congruency × Testing Session	-0.01	[-0.02, 0.00]	91.59%	0.014
			Temporal Proximity × Life Period Congruency × Testing Session	0.00	[-0.01, 0.01]	73.92%	0.007
Positive	Adjacent lag	.169	Intercept	-0.80	[-0.95, -0.65]	100.00%	1.10×10⁹
			Temporal Proximity	-0.51	[-0.64, -0.39]	100.00%	4.91×10⁶
			Life Period Congruency	0.27	[0.16, 0.39]	100.00%	690.04
			Testing Session	-0.08	[-0.22, 0.07]	86.29%	0.135
			Temporal Proximity × Life Period Congruency	-0.09	[-0.20, 0.02]	83.67%	0.189
			Temporal Proximity × Testing Session	0.07	[-0.05, 0.18]	88.37%	0.117
			Life Period Congruency × Testing Session	-0.03	[-0.14, 0.08]	72.40%	0.068
			Temporal Proximity × Life Period Congruency × Testing Session	-0.01	[-0.12, 0.10]	57.66%	0.057
Positive	Remote lags	.099	Intercept	-3.06	[-3.23, -2.89]	100.00%	1.14×10⁴⁸

			Temporal Proximity	-0.33	[-0.45, -0.20]	100.00%	1.78×10³
			Life Period Congruency	0.15	[0.02, 0.28]	98.88%	1.03
			Testing Session	0.01	[-0.15, 0.16]	53.01%	0.082
			Temporal Proximity × Life Period Congruency	0.05	[-0.08, 0.18]	78.44%	0.088
			Temporal Proximity × Testing Session	-0.01	[-0.14, 0.11]	59.19%	0.068
			Life Period Congruency × Testing Session	-0.09	[-0.22, 0.04]	91.98%	0.173
			Temporal Proximity × Life Period Congruency × Testing Session	0.07	[-0.06, 0.19]	85.75%	0.117
Negative	Power exponent	.454	Intercept	-0.12	[-0.13, -0.11]	100.00%	1.67×10²⁰
			Temporal Proximity	-0.04	[-0.05, -0.03]	100.00%	9.73×10⁵
			Life Period Congruency	0.06	[0.05, 0.07]	100.00%	1.77×10⁹
			Testing Session	-0.02	[-0.03, 0.00]	99.60%	0.203
			Temporal Proximity × Life Period Congruency	-0.00	[-0.01, 0.01]	65.63%	0.006
			Temporal Proximity × Testing Session	-0.01	[-0.02, 0.00]	91.56%	0.014
			Life Period Congruency × Testing Session	0.00	[-0.01, 0.02]	78.15%	0.007
			Temporal Proximity × Life Period Congruency × Testing Session	0.01	[0.00, 0.02]	88.02%	0.011
Negative	Adjacent lag	.108	Intercept	-0.81	[-0.97, -0.65]	100.00%	2.19×10⁸
			Temporal Proximity	0.13	[-0.01, 0.27]	97.12%	0.410
			Life Period Congruency	-0.29	[-0.42, -0.14]	99.99%	165.53
			Testing Session	-0.13	[-0.29, 0.03]	94.12%	0.262
			Temporal Proximity × Life Period Congruency	-0.19	[-0.33, -0.05]	99.55%	2.50

			Temporal Proximity × Testing Session	0.00	[-0.14, 0.15]	51.27%	0.071
			Life Period Congruency × Testing Session	0.07	[-0.08, 0.21]	82.22%	0.110
			Temporal Proximity × Life Period Congruency × Testing Session	0.09	[-0.06, 0.23]	89.13%	0.153
Negative	Remote lags	.296	Intercept	-2.41	[-2.52, -2.31]	100.00%	4.12×10⁵⁹
			Temporal Proximity	0.17	[0.09, 0.25]	100.00%	114.54
			Life Period Congruency	0.35	[0.26, 0.43]	100.00%	2.07×10⁷
			Testing Session	0.01	[-0.09, 0.11]	56.04%	0.052
			Temporal Proximity × Life Period Congruency	0.14	[0.06, 0.23]	99.96%	10.66
			Temporal Proximity × Testing Session	-0.04	[-0.12, 0.04]	83.76%	0.067
			Life Period Congruency × Testing Session	0.01	[-0.08, 0.09]	55.57%	0.042
			Temporal Proximity × Life Period Congruency × Testing Session	0.02	[-0.06, 0.10]	68.21%	0.046

Legend: R^2 = coefficient of determination, 95% HDI = 95% highest density interval, pd = probability of direction, Savage-Dickey BF = Savage-Dickey Bayes Factor. Fixed effects with a BF > 1 are bolded.

Table S3: Fit statistics and fixed-effects parameters for Bayesian mixed effects models related to intrusion probabilities based on semantic similarity

Measure	Bayes R^2	Fixed Effect	Estimate (b)	95% HDI	pd	Savage-Dickey BF
Slope	.114	Intercept	0.07	[0.07, 0.08]	100.00%	3.60×10^{38}
		Temporal Proximity	0.00	[0.00, 0.01]	91.10%	0.005
		Life Period Congruency	0.00	[0.00, 0.01]	88.75%	0.004
		Testing Session	-0.00	[-0.01, 0.00]	93.82%	0.007
		Temporal Proximity $\times$ Life Period Congruency	-0.00	[-0.01, 0.00]	81.47%	0.003
		Temporal Proximity $\times$ Testing Session	-0.00	[0.00, 0.00]	51.92%	0.002
		Life Period Congruency $\times$ Testing Session	-0.00	[-0.01, 0.00]	98.88%	0.026
		Temporal Proximity $\times$ Life Period Congruency $\times$ Testing Session	-0.00	[-0.01, 0.00]	88.42%	0.004

Legend: R^2 = coefficient of determination, 95% HDI = 95% highest density interval, pd = probability of direction, Savage-Dickey BF = Savage-Dickey Bayes Factor. Fixed effects with a BF > 1 are bolded.